

# Student Assignment Brief

## CONFIDENTIAL DOCUMENT

This document is intended solely for Softwarica College of IT & E-Commerce students for their own use in completing their assessed work for this module. It must not be passed to third parties or posted on any website.

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### Assignment Information

|                   |                              |
|-------------------|------------------------------|
| Module Name:      | Big Data Programming Project |
| Module Code:      | ST5011CEM                    |
| Assignment Title: | Coursework                   |
| Assignment Due:   | TBD                          |

|                           |                                                                   |
|---------------------------|-------------------------------------------------------------------|
| <b>Assignment Credit:</b> | 10 credits                                                        |
| <b>Word Count:</b>        | 2000 (excluding References and Appendix at the end of the report) |
| <b>Assignment Type:</b>   | Individual Coursework                                             |
| <b>Grading:</b>           | 67%                                                               |

### Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

### Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

## Assessed Module Learning Outcomes

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The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task.

This module helps students achieve the Course Learning Outcomes B1, B2, B4, B6, B7 and B8.

- B1: COMPUTATION THINKING: develop and understand algorithms to solve problems; measure and optimise algorithm complexity; appreciate the limits of what may be done algorithmically in reasonable time or at all.
- B2: PROGRAMMING: create working solutions to a variety of computational and real-world problems using multiple programming languages chosen as appropriate for the task.
- B4: DATA SCIENCE: work with (potentially large) datasets; using appropriate storage technology; applying statistical analysis to draw meaningful conclusions; and using modern machine learning tools to discover hidden patterns.

- B6: PROFESSIONAL PRACTICE: understand professional practices of the modern IT industry which include those technical (e.g. version control / automated testing) but also social, ethical & legal responsibilities.
  - B7: TRANSFERABLE SKILLS: apply a wide variety of degree level transferable skills including time management, team working, written and verbal presentation to both experts and non-experts, and critical reflection on own and others work.
  - B8: ADVANCED WORK: apply the above to advanced topics selected according to the interests of individual students.
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## Assignment Task

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### Background & Context

Urban transportation systems generate a vast amount of structured and semi-structured data — from bus timetables and live GPS tracking to route disruptions and fare information. Harnessing this data using data science and software engineering principles can lead to better decision-making, reduce delays, and improve the efficiency of public transport systems.

In this project, you will design, develop, and evaluate a data-driven predictive analytics solution using real-world or simulated data from sources such as the Bus Open Data Service (BODS) or similar.

### Introduction

This assessment requires you to complete an individual Big Data programming project that builds a Predictive Analytics Platform using large-scale transport data. You will demonstrate ability to integrate skills from the Advanced Algorithms (5003CEM) and Data Science (5005CEM) modules by performing comprehensive analysis, implementing machine learning, and delivering a complete data-driven solution.

**Key Technology Requirement:** PySpark is mandatory for large-scale distributed data processing, transformation, and machine learning tasks. Pandas and Dask are permitted and in some cases preferred for smaller preprocessing steps, exploratory analysis on sample data, and the final visualisation step (converting DataFrames to matplotlib / seaborn / plotly). You must justify your tool choice at each stage of the pipeline.

### Metric Definitions: Reliability & Efficiency

| Term                | Operational Definition                                                      | Applies To       | How Measured                                                             | Target / Threshold       |
|---------------------|-----------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------|--------------------------|
| Service Reliability | Proportion of journeys operating on schedule within an acceptable tolerance | Route / operator | % of trips arriving within $\pm 2$ min (urban) or $\pm 5$ min (rural) of | $\geq 85\%$ on-time rate |

|                         |                                                                       |                     |                                                                                                                        |                                          |
|-------------------------|-----------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|                         |                                                                       |                     | timetabled time                                                                                                        |                                          |
| Headway Regularity      | Consistency of interval between successive vehicles on the same route | Route / time-of-day | Standard deviation of observed headway; lower = more regular                                                           | SD $\leq$ 20% of scheduled headway       |
| Travel Time Variability | Spread of actual journey times relative to scheduled duration         | Trip / segment      | Coefficient of Variation (CV) of trip duration; lower = more reliable                                                  | CV $\leq$ 15%                            |
| Service Efficiency      | Ratio of scheduled capacity utilised to total capacity deployed       | Operator / fleet    | Passenger load $\div$ total seat capacity per trip (where data available); else trips completed $\div$ trips scheduled | Completion rate $\geq$ 95%               |
| Algorithmic Efficiency  | Runtime and resource use of data processing steps                     | Pipeline stage      | Wall-clock time (seconds); Spark UI stage metrics (task duration, shuffle size)                                        | Document baseline; justify optimisations |
| Model Efficiency        | Accuracy achieved relative to training cost                           | ML model            | F1-score (or RMSE) per second of training on held-out test set                                                         | Report for all 3+ compared models        |

**Note:** If your project uses a metric not listed above (e.g. a composite score), define it precisely in your Introduction and justify its validity against existing transport literature.

## Big Data Scale Requirement

A genuine big data pipeline must be demonstrated. Your analysis must be performed on a dataset of at least 100,000 records. If the BODS or similar datasets you ingest fall below this threshold, you must augment them using one or more of the following strategies:

1. **Multi-catalogue ingestion:** Combine multiple BODS catalogues (Timetables, Disruptions, Location, Fares). Document how datasets are joined and the combined row count after merging.
2. **Extended date range:** Ingest data covering at least a 3-month window to increase volume. Justify the period chosen.
3. **Synthetic data augmentation:** Use libraries such as Faker or SDV to synthesise additional rows preserving the statistical distribution of originals. Document the augmentation method and justify representativeness.
4. **Supplementary public datasets:** Add complementary open datasets (e.g. from data.gov.uk, OpenStreetMap, ONS travel statistics), citing all sources.

Regardless of file size, you must configure PySpark with at least 4 partitions, demonstrate caching and repartitioning, and include at least one Spark UI screenshot showing partition utilisation. If you use Pandas/Dask for a processing step, explicitly state why and compare the approach to a PySpark equivalent

# Core Requirement

## 1. Data Collection & Ingestion

- Select and use open datasets (e.g. data.gov.uk, BODS at <https://data.bus-data.dft.gov.uk/>).
- Ingest timetable, disruption, location, and/or fare data using Python (PySpark, Pandas, or Dask) or SQL.
- Clean and preprocess the data into a usable format for analysis.
- Ensure total record count meets the 100,000-record threshold (see Big Data Scale Requirement).

## 2. Data Storage & Processing

- Store datasets efficiently using relational databases (e.g. SQLite, PostgreSQL) or distributed DataFrames.
- **Tool guidance:** Use PySpark DataFrames for large-scale transformations and SQL queries. Pandas/Dask may be used for smaller aggregations, schema validation, or pre-visualisation reshaping — justify your choice in the report.
- Create relationships between datasets (e.g. linking timetables with disruptions by Service Code or Operator).
- Apply secure database connectivity: parameterised queries to prevent SQL injection.

## 3. Analytics & Prediction

- **Model requirement:** Implement at least one ML algorithm and compare a minimum of 3 models or configurations within your chosen category (classification, regression, or clustering).
- Predict timetable delays or non-compliance; detect disruption patterns; or forecast peak travel hours.
- Evaluate using the reliability/efficiency metrics defined in this brief (Section 4).
- Evaluate algorithmic complexity and accuracy (B1).
- Where PySpark MLlib is used, document the ML pipeline (VectorAssembler, stages, CrossValidator, etc.).

## 4. System Development

- Build the analytics pipeline as a coherent software solution.
- Include an architecture diagram: data ingestion → processing → model → visualisation.
- *Optional:* Create a simple dashboard (Streamlit, Dash, or Flask) to visualise outputs such as delay predictions, affected regions, or operator performance.

## 5. Security & Professional Practice

- Apply version control (Git/GitHub) — include a repository link in your submission.

- Apply secure data handling: parameterised queries, no hard-coded credentials.
- Document configuration (SparkSession settings, database connection strings) in a README or config file.
- Reflect on ethical, social, and legal considerations in using public transport data (GDPR, bias, data security).

## 6. Evaluation & Critical Reflection

- Conduct a thorough evaluation of your system's performance using the defined metrics.
- Reflect on technical choices, data limitations (including the memory-vs-distributed trade-off), challenges, and future improvements.
- Participate in an individual viva to present and defend your solution.

## Technical Requirement

### PySpark Operations (Required)

1. Load datasets using PySpark DataFrames.
2. Perform all large-scale data transformations using PySpark operations.
3. Use PySpark SQL for complex queries and aggregations.
4. Demonstrate understanding of distributed computing concepts (partitioning, lazy evaluation, DAG).
5. Optimise operations through caching, repartitioning, and broadcast joins where appropriate.
6. Report partition counts, stage metrics, and at least one Spark UI screenshot.

### Data Management

1. Database design and implementation for storing processed data.
2. Parameterised queries (no string concatenation) to prevent SQL injection.
3. Integration of multiple datasets using joins (PySpark or SQL).
4. Data persistence strategies: explain checkpointing or intermediate writes.

### Exploratory Data Analysis

1. Use PySpark DataFrame operations (describe, groupBy, agg) for large-scale exploration.
2. Calculate statistical measures (mean, median, std, skewness, kurtosis) using PySpark functions.
3. Perform data profiling and quality assessment (null counts, cardinality, outlier detection).
4. Create visualisations using matplotlib/seaborn/plotly — convert to Pandas only at the final plotting step.
5. Demonstrate data distributions, correlations, and patterns relevant to your business problem.

### Machine Learning (Required — compare at least 3 models)

- **Classification:** Evaluate with Accuracy, Precision, Recall, F1-score, ROC-AUC.

- **Regression:** Evaluate with RMSE, MAE,  $R^2$ . Use defined reliability/efficiency metrics as target or evaluation variable where applicable.
- **Clustering:** Evaluate with Silhouette Score and Within-Cluster Sum of Squares (WCSS).

## Project Topic Selection

You must identify a business stakeholder who would benefit from your analysis. Examples include:

- **Transport Authority / Local Council:** Monitoring operator compliance and service reliability.
- **Bus Operator:** Identifying routes with high delay frequency to optimise scheduling.
- **Commuter / Passenger App:** Predicting arrival times and disruptions in real time.
- **Urban Planner / Smart City Team:** Understanding peak travel patterns for infrastructure decisions.
- **Government Regulator:** Benchmarking operator performance against reliability thresholds.

## Example Project Ideas

1. Delay Prediction: Build a regression model to predict timetable delay in minutes; use Travel Time Variability as the target metric.
2. Non-Compliance Classification: Classify trips as compliant / non-compliant; evaluate with Service Reliability metric.
3. Disruption Pattern Detection: Cluster routes or time-periods by disruption frequency; report Headway Regularity.
4. Peak Hour Forecasting: Time-series or regression model to forecast passenger demand by hour/day.
5. Operator Performance Benchmarking: Compare operators on defined Service Efficiency and Reliability metrics.
6. Route Anomaly Detection: Identify unusual patterns in GPS location data indicative of unregistered diversions.

**Note:** Even if students use the same dataset, each project title and application must be different. Strict academic misconduct procedures apply.

## Report Structure

Your report must contain all sections below.

### Cover Page

Project Title; Student Name & ID; Module Name: Big Data Programming Project; Supervisor Name; Date of Submission.

### 1. Executive Summary (~150–200 words)

Brief overview of objectives, datasets used, techniques applied, key results, and outcomes.

## 2. Introduction

Problem Statement: What problem is being solved?

Purpose and Scope: What is the goal of the project?

Relevance: Why is this problem important in the context of smart cities?

Metric Definitions: State which reliability/efficiency definitions from Section 4 apply to your project and justify any custom metrics.

Learning Outcomes Targeted: B1–B8.

## 3. Literature Review / Background (Optional but encouraged)

Overview of related work in transport analytics or smart city data science.

Brief summary of existing predictive approaches in public transport systems.

Motivation for chosen methods and data.

## 4. Data Collection & Preprocessing

Data Sources: Which BODS catalogues (Timetables, Disruptions, Location, Fares) or supplementary datasets were used.

Data Scale: State total record count after any augmentation; confirm the 100,000-record threshold is met or explain your approach.

Tools & Technologies: Python (PySpark, Pandas, Dask), SQL, APIs — justify tool choice per stage.

Data Cleaning: Handling missing values, standardising formats.

Merging Strategy: How datasets were related and combined (e.g. by Service Code, Operator).

Challenges Encountered and how they were resolved.

## 5. Methodology

Feature Engineering: Variables selected/created for prediction.

Model Selection: Algorithms used (name all 3+ compared models); justify choices.

Data Splitting: Train/test split, cross-validation strategy.

Algorithm Complexity: Time/space analysis (Big-O) where applicable.

## 6. System Design and Implementation

Architecture Diagram: data ingestion → processing → model → visualisation.

Software Stack: Languages, database(s), libraries, version control.

Big Data Evidence: Spark UI screenshots, partition count, caching strategy.

Security Considerations: SQL injection prevention, data access handling.

User Interface (Optional): Dashboard or CLI for predictions.

## 7. Results and Evaluation

Model Performance Metrics: Report using the defined reliability/efficiency metrics; include Accuracy, Precision, Recall, F1-score, ROC curve, RMSE, Silhouette Score — as applicable.

Visualisations: Charts and graphs showing predictions, delay trends, operator comparison.

System Evaluation: Usability, performance, scalability.

Validation: How results were verified (against disruption records, operator scores, etc.).

## 8. Critical Reflection

What worked well?

Key challenges and how they were overcome.

Limitations of your approach (address the memory-vs-distributed trade-off).

Opportunities for improvement and future work.

Ethical, legal, or social considerations (GDPR, bias in data, accessibility).

## 9. Conclusion

Summarise achievements.

Highlight how the project met the learning outcomes.

Final thoughts on the usefulness and potential of the solution.

## 10. References

APA style referencing of all data sources, literature, and tools.

## 11. Appendices

Code snippets or pseudocode.

Sample datasets.

Extended visualisations or logs.

Screenshots of UI (if applicable).

GitHub repository link (if public).

# Submission Requirements

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### GitHub Repository must contain:

1. All Python code (.py files and/or .ipynb notebooks).
2. Spark/database configuration files.
3. README with detailed setup instructions.
4. requirements.txt.
5. Data preprocessing scripts (including augmentation script if applicable).
6. ML pipeline code.
7. Documentation folder.

8. Sample outputs/results.

#### Database Export:

- SQL dump or NoSQL export.
- Database schema diagram.
- Sample queries.

#### Technical Report (PDF format):

- All sections as outlined above.
- Screenshots of code execution.
- Spark UI screenshots (required — must show partition utilisation).
- Visualisations and results.
- All figures numbered and captioned.

#### Submission Instructions

| Requirement       | Details                                                                |
|-------------------|------------------------------------------------------------------------|
| File Naming       | NAME_studentID                                                         |
| File Format       | .pdf format                                                            |
| Submission Method | Campus 4.0 platform (submission link provided 2 weeks before deadline) |

## Marking and Feedback

#### How will my assignment be marked?

Your assignment will be marked by the Module Team using standardized criteria.

#### How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

### What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

### Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

## Assignment Support and Academic Integrity

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### Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

### Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

### Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**.

We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student handbook in Campus 4.0.

All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

### Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

### Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

### Administration of Assessment

|                             |                           |
|-----------------------------|---------------------------|
| <b>Module Leader Name:</b>  | Mr. Siddhartha Neupane    |
| <b>Module Leader Email:</b> | stw0084@softwarica.edu.np |
| <b>Assignment Category:</b> | Written                   |
| <b>Attempt Type:</b>        | Regular                   |
| <b>Component Code:</b>      | Coursework                |

## Assessment Criteria

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### Marking Criteria

| Assessment Component                                                       | Marks | Learning Outcomes Addressed |
|----------------------------------------------------------------------------|-------|-----------------------------|
| 1. Problem Definition, Planning, and Dataset Selection                     | 15    | B1, B4, B7                  |
| 2. Data Ingestion, Cleaning, and Preprocessing                             | 15    | B2, B4, B5                  |
| 3. Predictive Model Design and Implementation                              | 20    | B1, B2, B4, B8              |
| 4. System Development and Integration                                      | 15    | B2, B5, B6                  |
| 5. Output Presentation and Visualizations                                  | 10    | B4, B5, B7                  |
| 6. Evaluation and Critical Reflection (technical & ethical considerations) | 15    | B6, B7, B8                  |
| 7. Final Report Quality and Viva Presentation                              | 10    | B7, B8                      |
| Total                                                                      | 100   |                             |

### Marking Rubric

| Mark Range | Classification            | Performance Description – What the Student's Project Should Demonstrate                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90 – 100   | First Class (Exceptional) | <ul style="list-style-type: none"><li>- Original, innovative solution beyond expectations.</li><li>- Fully functional predictive system with clear architecture.</li><li>- Excellent integration of data sources (Timetables, Disruptions, Location, etc.).</li><li>- Model is accurate, well-evaluated, and explains variability in outcomes.</li><li>- Outstanding report: well-structured, insightful, critically reflective.</li><li>- Clear evidence of advanced understanding and technical proficiency.</li></ul> |
| 80 – 89    | First Class (Excellent)   | <ul style="list-style-type: none"><li>- Highly effective, complete system that meets all core requirements.</li><li>- Multiple data sources joined logically and used meaningfully.</li><li>- Strong predictive modelling with proper evaluation metrics.</li><li>- Strong software design, secure and modular code.</li><li>- Detailed documentation and thoughtful reflection.</li></ul>                                                                                                                               |
| 70 – 79    | First Class (Very Good)   | <ul style="list-style-type: none"><li>- Solid implementation with all major elements in place.</li><li>- Good use of multiple data sources with minor limitations.</li><li>- Predictive model is functional and tested but may have limited complexity.</li></ul>                                                                                                                                                                                                                                                        |

|                 |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                                 | <ul style="list-style-type: none"> <li>- Report covers all areas with clarity.</li> <li>- Evidence of good understanding of algorithms, data handling, and system development.</li> </ul>                                                                                                                                                                                                                      |
| <b>60 – 69</b>  | <b>Upper Second Class (2:1)</b> | <ul style="list-style-type: none"> <li>- Competent system with working components.</li> <li>- Basic integration of timetable/disruption/location data.</li> <li>- Model is implemented and evaluated, but with some limitations (e.g., lack of tuning or justification).</li> <li>- Report is complete but may lack depth in evaluation/reflection.</li> <li>- Basic attempt at ethical discussion.</li> </ul> |
| <b>50 – 59</b>  | <b>Second Class (2:2)</b>       | <ul style="list-style-type: none"> <li>- Partial implementation of the system.</li> <li>- Some data used but integration is weak or inconsistent.</li> <li>- Predictive analysis is limited in scope, accuracy, or relevance.</li> <li>- Report lacks structure, justification, or depth.</li> <li>- Minimal critical reflection.</li> </ul>                                                                   |
| <b>40 – 49</b>  | <b>Third Class</b>              | <ul style="list-style-type: none"> <li>- Basic or incomplete system, possibly not functional end-to-end.</li> <li>- Poor or no integration between datasets.</li> <li>- Predictive aspect is trivial or unclear.</li> <li>- Report is superficial with little technical justification.</li> <li>- Little to no reflection or evaluation.</li> </ul>                                                            |
| <b>Below 40</b> | <b>Fail</b>                     | <ul style="list-style-type: none"> <li>- Major components missing.</li> <li>- System not functional or lacks required data science/predictive element.</li> <li>- Inadequate or absent report.</li> <li>- No evidence of learning outcomes being met.</li> </ul>                                                                                                                                               |
| <b>0</b>        | <b>No Submission</b>            | <ul style="list-style-type: none"> <li>- No work submitted at all.</li> </ul>                                                                                                                                                                                                                                                                                                                                  |